

Soham Kharabe

C_C2_23

ML Lab - 1

```
In [1]: import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.preprocessing import StandardScaler

from sklearn.model_selection import train_test_split

import warnings
warnings.filterwarnings("ignore")
```

```
In [2]: df = pd.read_csv("titanic.csv")
```

```
In [3]: df.head()
```

Out[3]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500



In [4]:

```
df.tail()
```

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.00	
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.00	
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.45	
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.00	
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.75	



In [5]: `df.shape`

Out[5]: (891, 12)

In [6]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null    int64
1   Survived     891 non-null    int64
2   Pclass       891 non-null    int64
3   Name         891 non-null    object
4   Sex          891 non-null    object
5   Age         714 non-null    float64
6   SibSp        891 non-null    int64
7   Parch        891 non-null    int64
8   Ticket       891 non-null    object
9   Fare        891 non-null    float64
10  Cabin        204 non-null    object
11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [7]: `df.describe()`

Out[7]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200



In [8]: `df.isnull().sum()`

Out[8]:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

In [9]: `df["Age"].fillna(df["Age"].median(), inplace=True)`

In [10]: `df.isnull().sum()`

Out[10]:

PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

In [11]: `df["Embarked"].fillna(df["Embarked"].mode()[0], inplace=True)`

```
In [12]: df.isnull().sum()
```

```
Out[12]: PassengerId      0
Survived      0
Pclass      0
Name      0
Sex      0
Age      0
SibSp      0
Parch      0
Ticket      0
Fare      0
Cabin      687
Embarked      0
dtype: int64
```

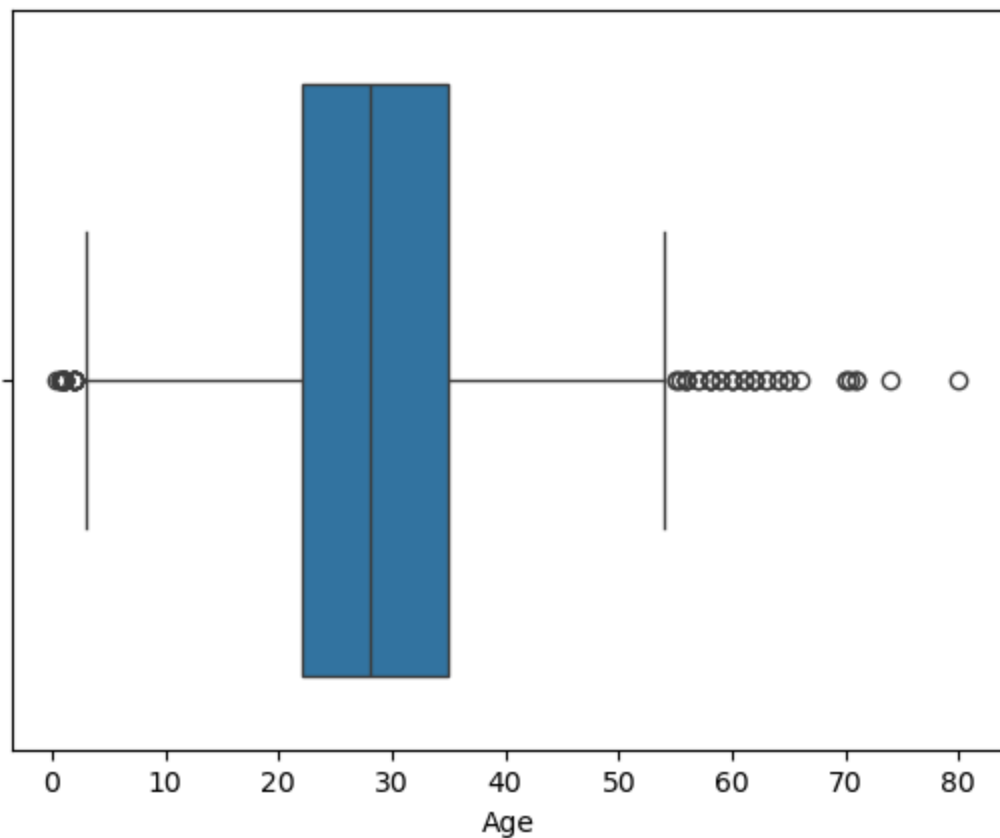
```
In [13]: df.drop("Cabin",axis=1,inplace = True)
```

```
In [14]: df.duplicated().sum()
```

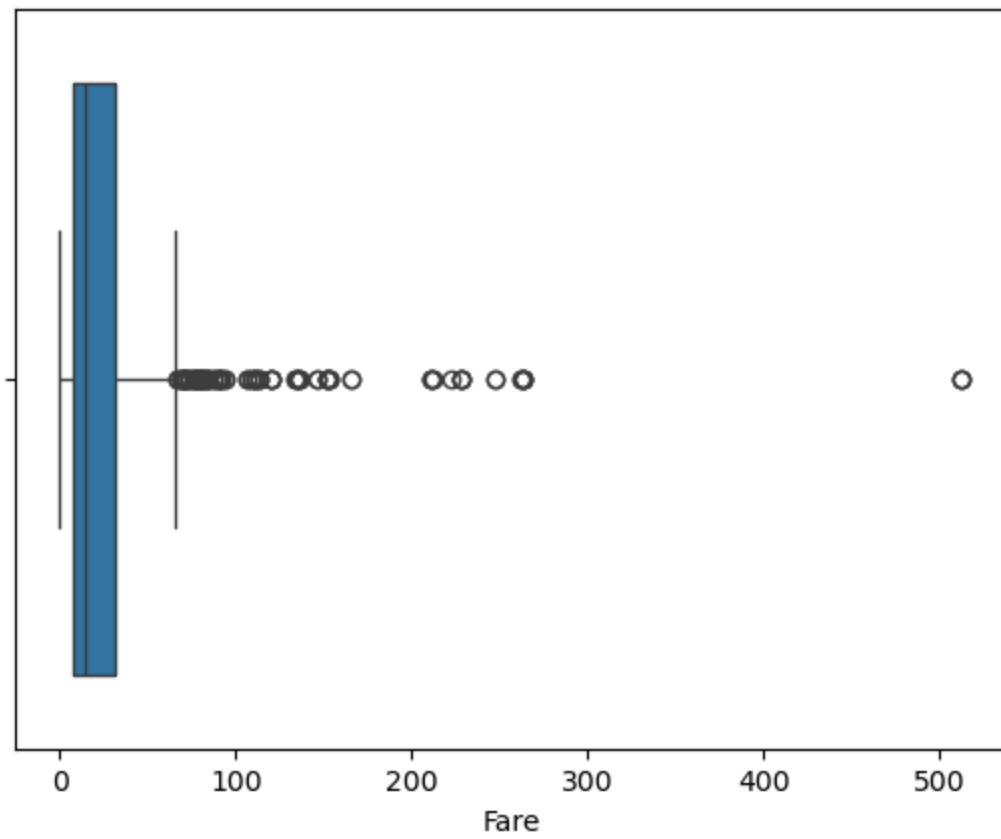
```
Out[14]: np.int64(0)
```

```
In [15]: df.drop_duplicates(inplace=True)
```

```
In [16]: sns.boxplot(x=df["Age"])
plt.show()
```

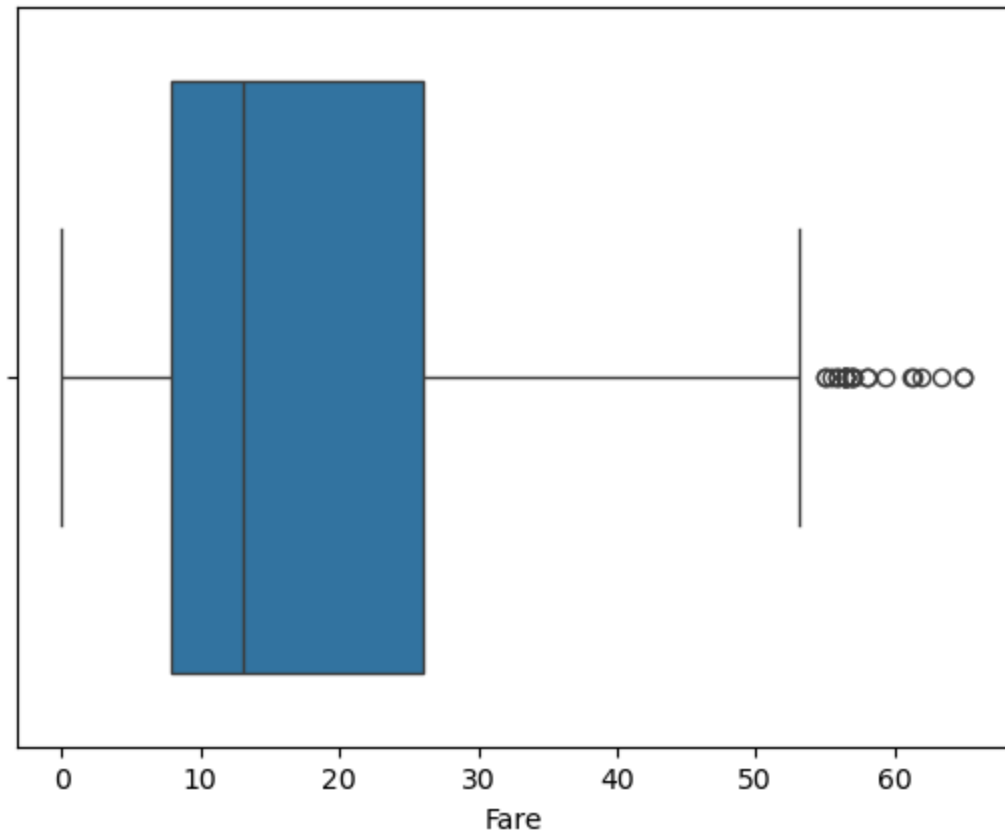


```
In [17]: sns.boxplot(x=df["Fare"])  
plt.show()
```



```
In [18]: Q1 = df["Fare"].quantile(0.25)  
Q3 = df["Fare"].quantile(0.75)  
  
IQR = Q3-Q1  
lower = Q1 - 1.5 * IQR  
upper = Q3 + 1.5 * IQR  
  
df = df[(df["Fare"] >= lower) & (df["Fare"] <= upper)]
```

```
In [19]: sns.boxplot(x=df["Fare"])  
plt.show()
```



```
In [20]: le = LabelEncoder()  
  
df["Sex"] = le.fit_transform(df["Sex"])
```

```
In [21]: df.head()
```

Out[21]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	I
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500	
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000	
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500	
5	6	0	3	Moran, Mr. James	1	28.0	0	0	330877	8.4583	

In [22]:

```
df = pd.get_dummies(df, columns=["Embarked"])
```

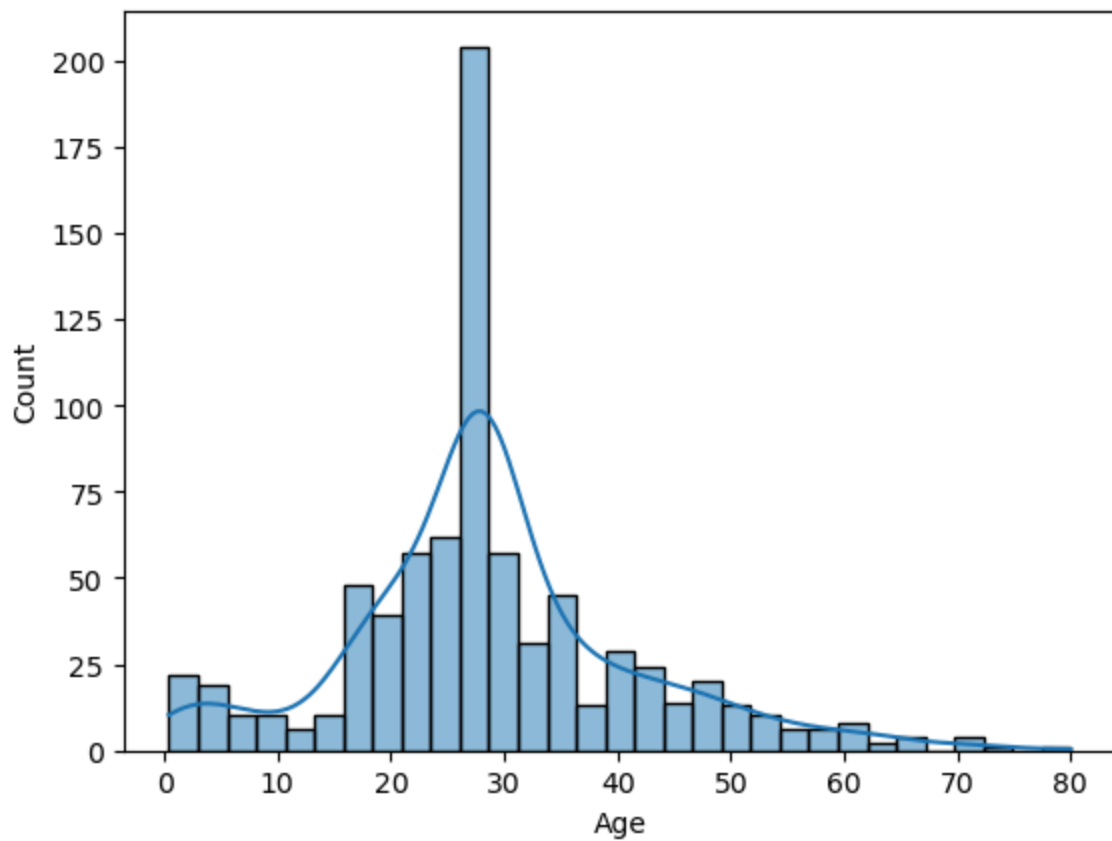
In [23]:

```
df.head()
```

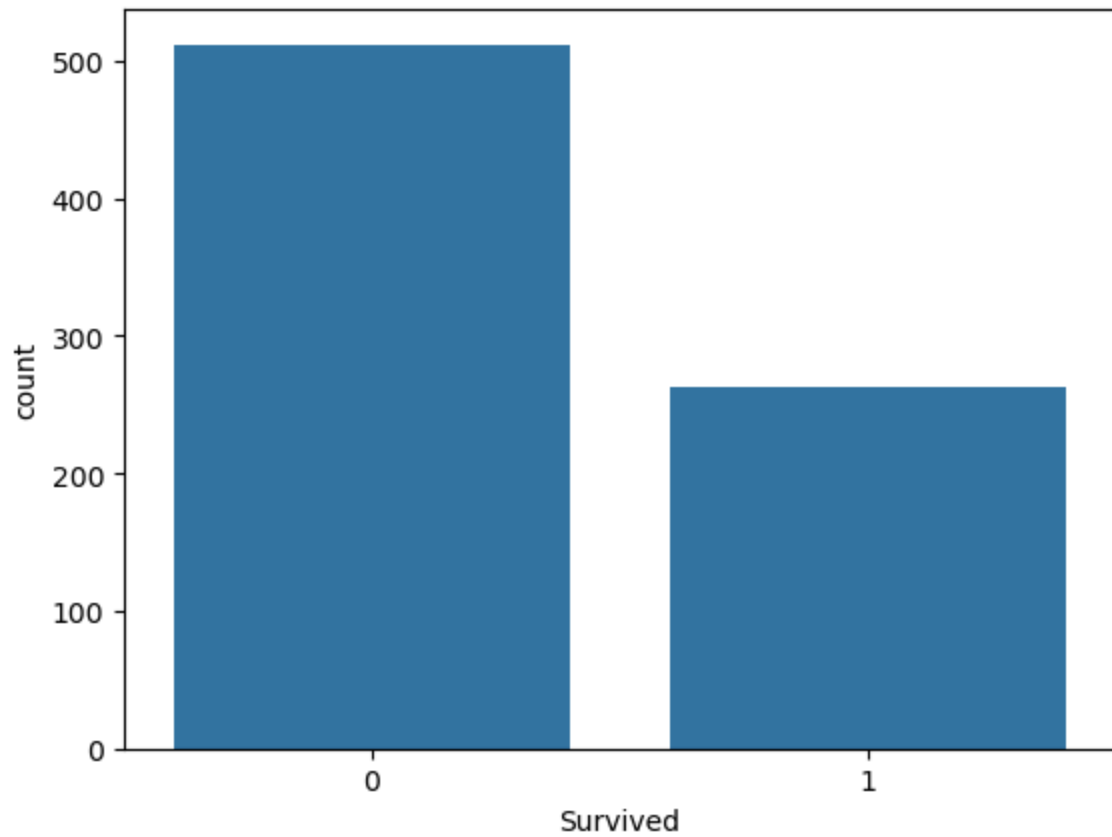
Out[23]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	I
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500	
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000	
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500	
5	6	0	3	Moran, Mr. James	1	28.0	0	0	330877	8.4583	

```
In [24]: sns.histplot(df["Age"], kde = True)  
plt.show()
```

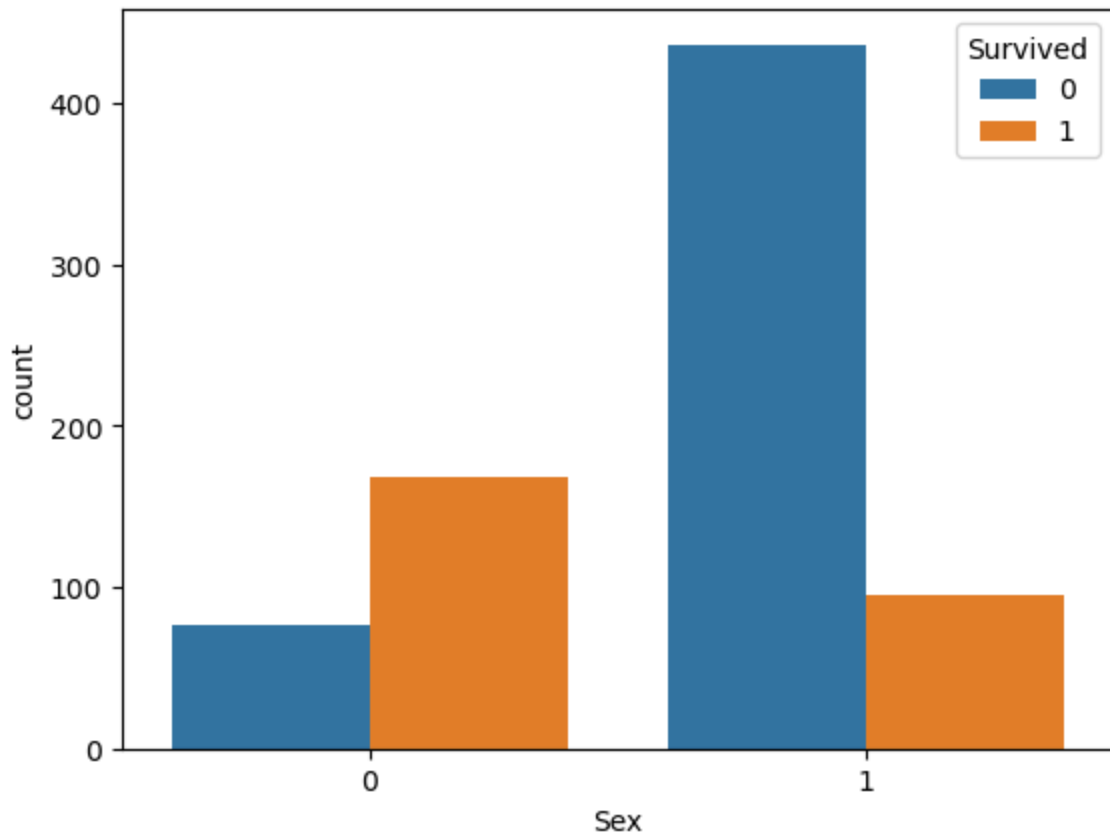


```
In [25]: sns.countplot(x="Survived", data=df)  
plt.show()
```



```
In [26]: sns.countplot(x="Sex",hue = "Survived",data=df)  
plt.show
```

```
Out[26]: <function matplotlib.pyplot.show(close=None, block=None)>
```

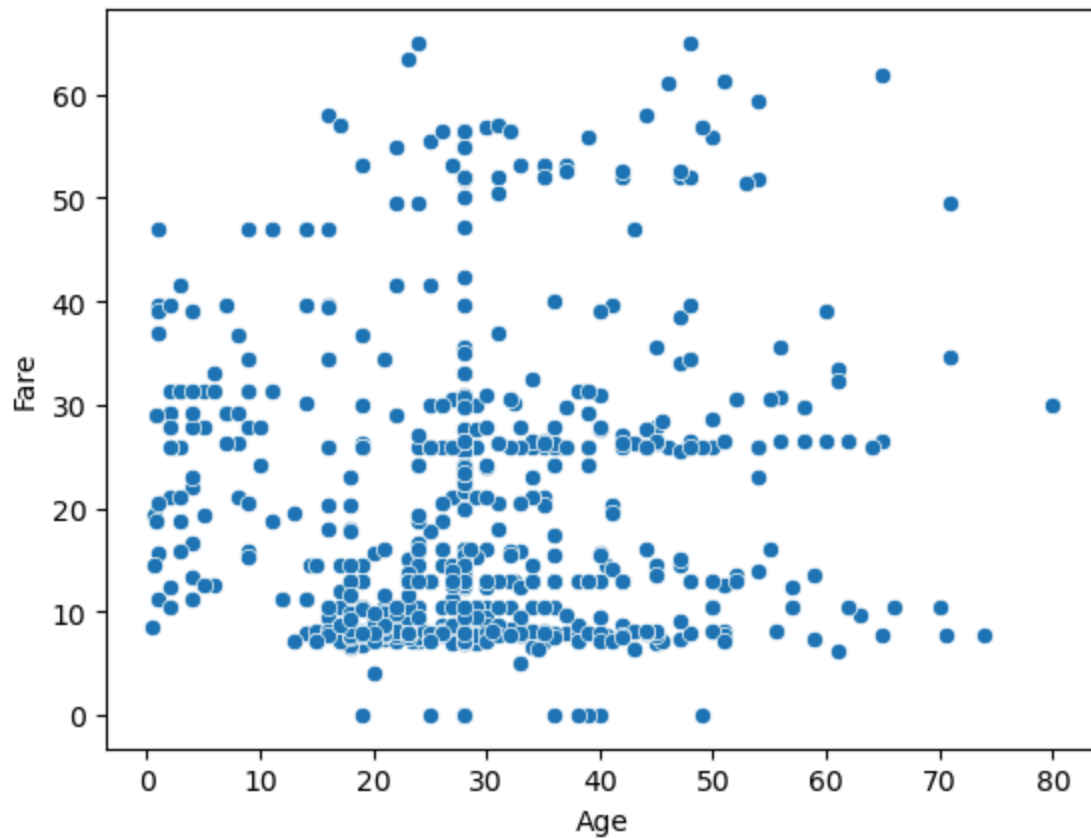


```
In [27]: df.head()
```

```
Out[27]:
```

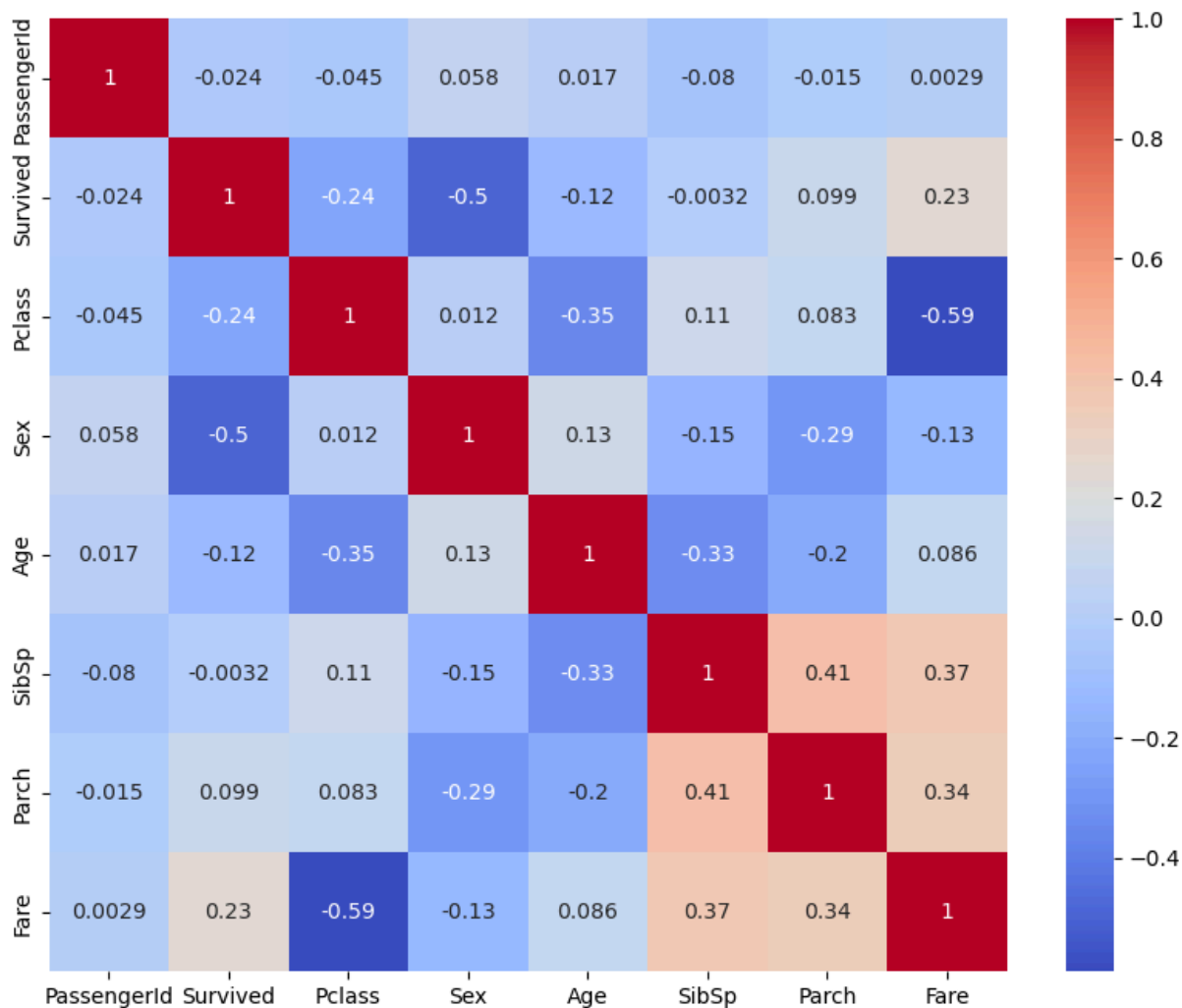
	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	1	22.0	1	0	A/5 21171	7.2500
2	3	1	3	Heikkinen, Miss. Laina	0	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	0	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	1	35.0	0	0	373450	8.0500
5	6	0	3	Moran, Mr. James	1	28.0	0	0	330877	8.4583

```
In [28]: sns.scatterplot(x="Age",y="Fare",data=df)
plt.show()
```



```
In [29]: plt.figure(figsize=(10,8))
numeric_df = df.select_dtypes(include=['number'])

sns.heatmap(numeric_df.corr(),
            annot = True,
            cmap = 'coolwarm')
plt.show()
```



```
In [30]: from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
df[["Age", "Fare"]] = scaler.fit_transform(df[["Age", "Fare"]])
```

```
In [31]: X = df.drop("Survived", axis = 1)
y = df["Survived"]
```

```
In [32]: X = X.drop(["PassengerId", "Name", "Ticket"], axis = 1)
```

```
In [33]: from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size = 0.20,
    random_state = 42
)
```

```
In [34]: print ("Training Data :", X_train.shape)
print ("Training Data :", X_test.shape)
```

Training Data : (620, 9)
Training Data : (155, 9)

```
In [35]: print (X_train.head())
```

	Pclass	Sex	Age	SibSp	Parch	Fare	Embarked_C	Embarked_Q	\
147	3	0	-1.546024	2	2	1.219877	False	False	
868	3	1	-0.058613	0	0	-0.613302	False	False	
60	3	1	-0.528321	0	0	-0.780650	True	False	
468	3	1	-0.058613	0	0	-0.744112	False	True	
777	3	0	-1.859163	0	0	-0.394057	False	False	

	Embarked_S
147	True
868	True
60	False
468	False
777	True